Use Cases

for

Student Management System, Release 1.0

Version 1.0 approved

Prepared by Team 1

Process Impact

March 30, 2026

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Revision History

Name	Date	Reason For Changes	Version
Đỗ Đăng Khoa, Cao Tiến Phát, Trần Việt Long	10/03/2026	Project Kick-off: Brainstormed and identified primary actors and the initial 15 use cases for the system.	1.0 draft 1
Đỗ Đăng Khoa	11/03/2026	Initial draft: Created document template and drafted normal flows for *UC-1 Register for Courses*	1.0 draft 1
Trần Việt Long	13/03/2026	Added detailed flows and exceptions for *UC-7 Input Midterm/Final Grades*	1.0 draft 2
Cao Tiến Phát	15/03/2026	Drafted *UC-2 View Timetable*; Integrated BR-11 for privacy constraints	1.0 draft 3
Đỗ Đăng Khoa	18/03/2026	Added *UC-10 Open Course Sections*; Refined Redis locking mechanisms and pre-conditions for UC-1 & UC-10	1.0 draft 4
Đỗ Đăng Khoa, Cao Tiến Phát, Trần Việt Long	20/03/2026	Baseline following group review and final inspection; Ready for Release 1.0	1.0 approved

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The various user classes identified the following primary actors and use cases for the Student Management System:

Primary Actor	***Use Cases***
Student	1. Register for Courses 2. View Timetable 3. Check Grades 4. Pay Tuition Fees 5. Submit Academic Request (e.g., leave of absence, grade appeal)
Lecturer	6. View Class Roster 7. Input Midterm/Final Grades 8. Take Attendance 9. Send Academic Communication
Academic Admin	10. Open Course Sections 11. Schedule Timetables 12. Manage Core Curriculum 13. Update Student Academic Status
Academic Advisor	14. Review Student Profile 15. Process Academic Request (Approve/Reject)

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ID and Name:	**UC-1 Register for Courses**		
Created By:	Do Dang Khoa	Date Created:	11/03/2026
Primary Actor:	Student	Secondary Actors:	Financial System, Training Database
Description:	A student accesses the system to select and register for class sections for the upcoming semester. The system validates prerequisites, credit limits, and tuition status before confirmation.		
Trigger:	The Student selects the "Course Registration" function from the main dashboard.		
Preconditions:	PRE-1: Student is logged in with a unique ID (per BR-13). PRE-2: The registration period is currently open (per BR-4).		
Postconditions:	POST-1. The Student is enrolled in the selected class sections. POST-2. The remaining seat capacity for each section is updated in the database.		
Normal Flow:	1. The Student requests to view available classes. 2. The System checks the Accounting Dept database for any outstanding tuition debt (BR-6). 3. The System displays the course catalog filtered by the Student's major. 4. The Student selects a class section to add. 5. The System verifies that the Student has successfully passed all prerequisite courses for that class (BR-1). 6. The System checks if the selected class time overlaps with any of the Student's already enrolled classes (BR-5). 7. The System attempts to acquire a Redis distributed lock to ensure the class has not exceeded seat capacity during the transaction (BR-3). 8. The System calculates the total credits currently selected; if adding this course keeps the total at or below 24 credits, the System adds the course (BR-2). 9. The System confirms successful registration.		
Alternative Flows:	4. Waitlist: If a class is full but allows a waitlist, the System asks the Student if they wish to be placed on the waitlist. If the Student confirms, the System adds them to the queue.		
Exceptions:	2.E1 Outstanding Debt: The System displays a "Registration Blocked" message due to debt and terminates the use case (BR-6). 5.E1 Prerequisite Missing: The System displays: "Registration denied: Prerequisite courses not completed." 6.E1 Schedule Conflict: The System displays: "Time conflict detected with [Course Name]." 7.E1 Class Full: The System displays: "Class capacity reached. Enrollment locked." 8.E1 Over Credit Limit: The System prevents the addition and displays: "Maximum limit of 24 credits exceeded."		
Priority:	High		
Frequency of Use:	Extremely high during the 2-week registration window at the start of each semester (approx. 5,000 students accessing the module multiple times).		
Business Rules:	BR-1, BR-2, BR-3, BR-4, BR-5, BR-6, BR-13.		

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Other Information:	1. The system must handle high concurrency.
Assumptions:	The response time for scheduling conflict checks (BR-5) and prerequisite checks (BR-1) must be under 2 seconds to avoid system congestion.

ID and Name:	**UC-7 Input Midterm/Final Grades**		
Created By:	Tran Viet Long	Date Created:	13/03/2026
Primary Actor:	Lecturer	Secondary Actors:	Financial System, Training Database
Description:	A Lecturer accesses the SMS to input, review, and officially submit midterm, process, and final exam grades for students in a specific course section they are assigned to teach.		
Trigger:	Lecturer navigates to the "Grade Management" module and selects a specific course section.		
Preconditions:	**PRE-1.** Lecturer is logged into the SMS and assigned to the selected course section. **PRE-2.** The current date is within the official grading window (exactly 7 working days after the final exam date).		
Postconditions:	**POST-1.** Student grades are securely stored in the SMS database. **POST-2.** Final grades are automatically calculated and letter grades are assigned. **POST-3.** The grading status of the course section is marked as "Submitted".		
Normal Flow:	**7.0 Manual Grade Entry and Submission** 1. Lecturer selects a course section from their assigned teaching list. 2. SMS displays the grading sheet, listing all enrolled students and input columns (Process, Midterm, Final Exam). 3. Lecturer manually inputs numeric grades (0.0 to 10.0) for each student. 4. SMS validates the inputted data in real-time. *(see 7.0.E1)* 5. Lecturer clicks "Calculate Final Grades". 6. SMS automatically calculates the numeric final grade based on the weighted formula (BR-8) and maps it to the standard letter grade.7. Lecturer clicks "Submit Official Grades". 8. SMS displays a warning prompt: *"Once submitted, grades cannot be modified without Academic Admin approval. Do you want to proceed?"* 9. Lecturer confirms the submission. 10. SMS saves the grades securely, updates the students' academic records, and sends a notification to the students that their grades are available.		
Alternative Flows:	**7.1 Upload Grades via Excel File** 1. In step 3 of the normal flow, Lecturer clicks "Import from Excel". 2. Lecturer uploads a pre-formatted Excel file containing the grades. 3. SMS parses the file and populates the grading grid on the screen. *(see 7.1.E1)* 4. Return to step 4 of the normal flow. **7.2 Save as Draft** 1. In step 5 of the normal flow, Lecturer clicks "Save Draft" instead of		

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	calculating/submitting. 2. SMS saves the current inputted values without applying formulas or finalizing. 3. SMS terminates the use case (Lecturer can return later).
Exceptions:	**7.0.E1 Invalid Grade Format (BR-7)** 1. SMS detects a grade input that is outside the 0.0 - 10.0 range or contains invalid characters (e.g., text). 2. SMS highlights the specific cell in red and displays an error tooltip. 3. SMS disables the "Calculate/Submit" buttons until the Lecturer corrects the error. 4. Return to step 3 of normal flow. **7.0.E2 Grading Period Closed** 1. Upon triggering the use case, SMS detects that more than 7 working days have passed since the final exam. 2. SMS displays the grading sheet in "Read-Only" mode. 3. SMS informs the Lecturer that the grading period is closed and provides a link to "Request Admin Override". 4. SMS terminates the use case. **7.1.E1 Excel File Parsing Error** 1. SMS detects that the uploaded Excel file has incorrect column headers, corrupted data, or unmatched Student IDs. 2. SMS rejects the file upload and displays a detailed error message indicating the problematic rows. 3. Lecturer fixes the Excel file and re-uploads, OR chooses to input manually.
Priority:	High
Frequency of Use:	Spikes dramatically at the end of each semester during the 7-day grading window.
Business Rules:	BR-7, BR-8, BR-9
Other Information:	High data integrity is required. Every modification to a grade after the official submission must be logged in an Audit Trail with the Academic Admin's approval (BR-9).
Assumptions:	Assume Lecturers have a stable internet connection during the final submission step to prevent partial data saving.

ID and Name:	**UC-2 View Timetable**		
Created By:	Cao Tien Phat	Date Created:	15/03/2026

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Primary Actor:	Student / Lecturer	Secondary Actors:	Training Database
Description:	A user (Student or Lecturer) accesses the system to view their personal weekly or semesterly class schedule. The system ensures users can only view their own timetable based on their enrollment or teaching assignments.		
Trigger:	The User clicks on the "Timetable" or "Schedule" module from the main navigation menu.		
Preconditions:	**PRE-1:** The User is successfully logged into the system with a unique ID (BR-13). **PRE-2:** The academic semester schedule has been officially published by the Academic Admin.		
Postconditions:	POST-1: The system successfully renders the correct personal timetable for the requested user on the screen.		
Normal Flow:	1. The User requests to view their timetable for the current week/semester. 2. The System identifies the User's Role (Student or Lecturer) and their unique ID. 3. The System queries the database for classes associated with the User's ID (enrolled classes for Students, assigned teaching classes for Lecturers). 4. The System enforces privacy policies, ensuring the retrieved data strictly belongs to the requesting User (BR-11). 5. The System formats the retrieved class times, room numbers, and instructor names into a graphical calendar interface. 6. The System displays the personalized timetable to the User.		
Alternative Flows:	**1a. Filter by Timeframe:** The User selects a different semester or a specific week from a dropdown menu. The System re-queries and updates the displayed timetable accordingly. **6a. Export/Sync:** The User clicks "Export to PDF" or "Sync to Calendar". The System generates a downloadable PDF file or a standard ICS calendar file.		
Exceptions:	**3.E1 Empty Schedule:** If the Student has not registered for any classes, or the Lecturer is not assigned to teach any classes this semester, the System displays: "No classes scheduled for the selected period." **4.E1 Unauthorized Access Attempt:** If a user attempts to view another person's schedule (e.g., by modifying URL parameters), the System blocks the request, logs the attempt, and displays: "Error 403: You are not authorized to view this timetable" (BR-11).		
Priority:	Medium		
Frequency of Use:	High (Accessed daily or weekly by almost all users during the active semester).		
Business Rules:	BR-11, BR-13		
Other Information:	The timetable UI must be highly responsive and display correctly on mobile devices, as students frequently check their schedules on smartphones while on campus.		
Assumptions:	It is assumed that the course schedule (including room assignments and timeslots) has already been created and verified for conflicts by the Admin.		

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ID and Name:	**UC-10 Open Course Sections**		
Created By:	Do Dang Khoa	Date Created:	18/03/2026
Primary Actor:	Academic Admin	Secondary Actors:	Training Database, Facility Management System
Description:	The Admin sets up specific class sections for a given semester. This involves assigning instructors, defining schedules (time and room), and setting the maximum seat capacity for each section.		
Trigger:	The Admin selects "Semester Management" and clicks "Open New Course Section."		
Preconditions:	PRE-1: The Admin is logged into the system with appropriate administrative privileges (BR-12). PRE-2: The master course catalog (list of all possible subjects) has been updated and exists in the database. PRE-3: The list of available instructors and their IDs is current (BR-13).		
Postconditions:	POST-1 New class sections are created and visible for student registration. POST-2 Instructor schedules are updated to reflect new assignments.		
Normal Flow:	2. The Admin selects the target Semester and the Subject to be opened. 3. The System displays the "Create Section" form. 4. The Admin enters a unique Section ID and assigns an Instructor from the staff list. 5. The Admin defines the schedule (Days of the week, Period, and Classroom). 6. The Admin enters the Maximum Seat Capacity for the section. 7. The System verifies that the Instructor does not have a conflicting class at the same time. 8. The System verifies that the selected Classroom is available at the specified time. 9. The Admin clicks "Save and Publish." 10. The System initializes the Redis distributed lock for this section to prepare for high-concurrency seat counting during registration (BR-3).		
Alternative Flows:	Step 8a. Save as Draft: The Admin chooses to save the section as "Draft" instead of "Published." The section is saved but remains invisible to students until the Admin changes the status to "Published."		
Exceptions:	6.0.E1 Instructor Conflict: The System displays: "Assignment Error: Instructor is already assigned to another class at this time." 7.0.E1 Room Conflict: The System displays: "Facility Error: Classroom is already occupied."		
Priority:	High		
Frequency of Use:	Low. Typically performed in bulk 2–4 weeks before a new semester begins (approximately 200–500 sections created per term).		
Business Rules:	BR-3, BR-12, BR-13		
Other Information:	If a section is modified after students have already registered, the system must trigger an automated email notification to all enrolled students. All database transactions involving section creation are logged for audit		

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	purposes.
Assumptions:	AS-1: The list of available classrooms provided by the Facility Management System is 100% accurate. AS-2: Course credits are already defined in the master catalog to ensure BR-2 (credit limits) can be calculated correctly later.

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